

Bullet Farm: Architecture and Pre-Release Inventory of a Software-Change Transaction Processor

Bullet Farm Maintainers

Abstract—Coding agents are proliferating as chat loops, plugin kernels, and meta-wrappers around existing CLIs. The scarce resource is not another persona or sandbox flag. It is a correct concurrent *software-change transaction*: who may mutate an exact repository subject, which incarnation owns that write, what evidence names that subject, and whether an ambiguous remote effect may be retried. Bullet Farm is a local-first transaction processor for software changes and model cognition in which models may propose typed patches but cannot grant write, verification, or merge authority. This preprint states that architecture, then inventories exactly what exists on 2026-08-25 against the family’s own evidence ladder (component, synthetic, transaction, live, release): every implemented boundary reported here is a commit-bound *component* proof; no transaction, live, or release proof exists; all 26 family release gates are blocked. It states a threat model that separates implemented boundaries from designed ones, describes precisely what two bounded TLC models do and do not check, presents evolutionary multi-agent control as a design rather than a result, and compares the authority contract—not performance—with Gas Town / Gas City, DeepSeek Harness, Omnigent, and single-agent harnesses at pinned commits. A reproducibility appendix lists the commands and outputs behind every number, and a red-team appendix answers or concedes the ten strongest attacks on the paper’s claims.

Index Terms—Multi-agent software engineering, transaction processing, Git isolation, evidence binding, authority fencing, coding agents, verification, model checking.

I. INTRODUCTION

The last two years produced a crowded market of coding agents: protocol-grade CLIs, IDE wrappers, hosted pull-request bots, research SWE loops, multi-agent orchestrators, plugin kernels, and meta-harnesses that wrap the previous categories [1]–[7]. Those systems differ in user experience and in how much of the model loop they own. They largely agree on an implicit completion rule: if a session looks done, a terminal is idle, a process exited zero, or a pull request opened, the work is complete.

That rule fails under concurrency. Public Gas Town and Gas City issues show the same family of faults repeating under load: two agents share one working tree and clobber uncommitted edits [8]; a timeout after a server-side commit mints duplicate live workflows and three writers enter one checkout [9]; a background `git checkout` against the town-root `.git` deletes operational files [10]; a session that died mid-task leaves its work claimed with no signal [11]; a patrol step classifies every live worker as dead immediately

before a forced worktree removal [12]. Those reports are not a verdict that Gas Town is a weak product. They are evidence that a role/session orchestrator is the wrong abstraction for adversarial multi-writer Git [13].

DeepSeek Harness and Omnigent improve different layers. DeepSeek treats every capability—model, tools, sandbox, session, and UI—as a Cordis plugin and makes the append-only session log the source of model-visible context [14]. Omnigent sits *above* Claude Code, Codex, Cursor, and other CLIs, adding OS sandboxes, an L7 credential proxy, contextual policies, and live collaboration [7], [15], [16]. Both are real advances. Neither binds write authority to an incarnation-fenced Attempt, evidence to an exact Candidate, or remote integration to a reconciled effect receipt. A plugin swap or a wrapped harness can still let the model run `git`.

Bullet Farm inverts the authority model [17], [18]. The designed law is compact:

Many agents may reason. Multiple isolated Variants may compete. Exactly one incarnation-fenced Attempt may write within each Variant. Every consequential external mutation is brokered. Every proof names its exact subject. The repository remains sovereign.

The contribution of this preprint is fivefold, and each item is bounded by the evidence class it carries (Section II):

- 1) It states that law as a five-plane transaction with typed subjects, not as a prompt convention (Sections V and IX) [DESIGN].
- 2) It reduces the public Gas Town / Gas City issue corpus and the Git documentation to a closed list of missing *bindings*, and maps each to a rule and to an honest implementation status (Sections IV and XIV) [OBSERVED].
- 3) It reports *exactly* which components have commit-bound receipts as of the 2026-08-25 reconciliation and which product gates are blocked, so a reader cannot confuse a green crate test with a release (Section X) [COMPONENT] [BLOCKED].
- 4) It states a threat model that lists implemented boundaries separately from designed ones, describes what two bounded TLC models encode and omit, and presents evolutionary control as design (Sections VI, VII, and XI).
- 5) It gives a falsification section, a reproducibility appendix with commands and observed outputs, and a red-team appendix that answers or concedes the ten strongest attacks on its own claims (Section XV, Appendices B and C).

This paper does not claim that Bullet Farm is faster, cheaper,

Placeholder authorship for this preprint. Named authors will be supplied before an arXiv upload. Affiliation: the Bullet Farm split family (public hub, kernel, BulletGit, and operations portal). This document is not a release receipt, installer, or benchmark result.

TABLE I
EVIDENCE CLASSES FROM THE RELEASE CONTRACT [23]. NOTHING IN
THIS PAPER IS ABOVE THE FIRST ROW.

Class	Proves	Does not prove
COMPONENT_PROOF	one crate, service, or portal surface passed its mapped tests	cross-process transaction safety
SYNTHETIC_PROOF	deterministic simulator behavior	a provider, forge, or production mutation
TRANSACTION_PROOF	one exact offline five-plane transaction with independent receipts	external provider or forge conformance
LIVE_PROOF	an admitted provider or effect adapter passed the same exact-subject transaction	a release on every platform
RELEASE_PROOF	packages, installer, recovery, security, signatures, provenance, live profiles from tagged bytes	future versions or untested environments

safer, or more autonomous than any named system. No public comparison is valid until the same receipt-bearing corpus runs on pinned upstream commits [19]. The family’s own historical adjudication scored the *paper* architecture 96.4/100 and its implementation maturity about 0.5/10 [20], [21]; an earlier, separately produced draft is recorded but not distributed [22]. The maturity number is now too low for many components and still correct for the connected product.

II. READING RULES AND EVIDENCE CLASSES

The family release contract defines five evidence classes [23], reproduced in Table I. This paper never promotes a claim above its class. Every empirical sentence is readable against the ladder through a bracketed tag: [COMPONENT] for a crate, service, or portal surface that passed its mapped tests at a named commit; [SYNTHETIC] for deterministic simulator behavior; [TRANSACTION], [LIVE], and [RELEASE] for the three classes that do not yet exist in this family; [DESIGN] for a rule that is specified in an accepted document and has no live code path; [ARCHIVED] for the one archived live demonstration, which sits outside the ladder; [OBSERVED] for a dated observation of an external system or of a file in the tree; and [BLOCKED] for a family release gate that the read-only inventory reports as blocked.

Three reading rules follow. First, a COMPLETE row in the closure plan [24] is always [COMPONENT]; the vocabulary means “an exact committed subject and a mapped receipt exist for this bounded claim,” nothing more. Second, an exit code, a model statement, an HTTP 2xx, a branch push, or a pull request is never evidence by itself, and UNKNOWN, timeout, zero tests, unsupported, skipped, or flaky never equals VERIFIED [23], [25]. Third, a newer commit invalidates a receipt until the mapped gate is replayed; the subjects this paper reports are those in Appendix A, and the working tree that compiled this PDF may be newer.

III. A TAXONOMY OF CODING AGENTS

Treating “coding agents” as one category hides the authority question. We distinguish six families by *who writes the repository* and *what signal is allowed to mean done*.

Everything in this section is [OBSERVED]: dated observation of public documentation, pinned in Appendix A.

A. Protocol-Grade Single Agents

Claude Code (bidirectional stream JSON), Codex (App Server JSON-RPC over stdio JSONL), Cursor (Agent Client Protocol), and Google Antigravity (agy structured headless mode) are the four V1 adapter targets [1]–[4], [18], [26]. On 2026-08-25, environment-cleared local `-version/-help` probes of absolute binaries observed Claude Code 2.1.243, Codex CLI 0.149.1, Cursor Agent 2026.08.11-e8db854, and agy 1.1.19, and recorded each executable’s SHA-256 [18]. Those observations are inventory, not certification. Cursor’s required ACP entry point remained UNKNOWN on that host; print/stream-JSON modes must not be substituted [18]. Each of these products is a capable single-agent harness. The model, or a tool the model invoked, writes the workspace. A “plan” flag or a vendor sandbox is defense in depth, not a foreign write gateway.

B. IDE and CLI Wrappers

Aider, Cline, Continue, Goose, and Roo Code wrap an editor or terminal around one or more models [27]–[31]. They differ in UX, MCP adoption [32], and how much Git they automate. They share the user’s checkout as the writable world. Isolation, if any, is a worktree, a subdirectory, or an honor-system ignore list.

C. Research SWE Loops

OpenHands, SWE-agent, Agentless, and AutoCodeRover optimize for SWE-bench-style issue resolution [33]–[38]. Their contribution is the agent–computer interface and the evaluation harness. Completion is a resolved issue or a passing hidden test, not a protected integration onto a sovereign default branch under an independent attestor.

D. Hosted Change Agents

GitHub Copilot’s coding agent, Jules, Devin, and Kiro accept a task and return a pull request or a hosted workspace [39]–[42]. The PR is the product. That is a legitimate remote-collaboration contract. It is not an incarnation-fenced local transaction: retries, reviewer independence, and evidence staleness after rebase are delegated to the host’s merge queue and to human review.

E. Orchestrators

Gas Town is a Go workspace manager for many Claude, Codex, Copilot, Gemini, or Cursor agents [5], [43]. Its README requires tmux 3.0+ for `gt up` and the tmux-backed roles, Beads (bd) 0.57.0+, Dolt, and ICU4C development headers for source builds; it persists work in git-backed *hooks* (worktrees), coordinates via Beads/Dolt, and lands work through a bisecting merge queue called the Refinery [5], [44]. Roles—Mayor, Witness, Deacon, Dogs, polecat, Refinery—are operationally legible. Additional product machinery is

easy to underestimate: convoys bundle beads; molecules instantiate formulas; a watchdog hierarchy (Witness, Deacon, Dogs) patrols health; a scheduler caps polecat dispatch (`max_polecats`); Seance recovers predecessor session logs; Wasteland federates towns [5]. Gas City generalizes the same factory into Agent, Bead, Formula, Rig, Pack, and Event; the dated competitor snapshot records typed run/session APIs, warm projections, formula-v2 retry/fan-out/drain controls, degraded storage outcomes, and release artifacts with checksums, SBOMs, and attestations in v1.4.1 [19], [45], [46]. The product leverage is real. The authority model is not: liveness is a terminal multiplexer, isolation is a worktree or an optional pack script, and the fleet can be the merge authority. Bullet Farm should be measured against those product properties—durable work, role flexibility, fan-out receipts, provider plurality, health reconciliation, and release evidence—not against a cartoon of “tmux and vibes” [19].

F. Plugin Kernels

DeepSeek Harness (`dsh`) is a developer-preview agent runtime whose doctrine is “everything is a plugin” [6], [47]. Cordis mounts model adapters, tools, sandboxes, storage, loops, scheduling, and the UI into one context [14], [48]. A running instance is a plugin tree composed from bundles and ordered patches. The session log is the source of model-visible context: anything the model sees must be reconstructable from that log. A turn is zero or more steps; each step is one model request plus the tools it calls; `agent/pre-step` may rewrite or reject claimed messages; tool execution is a `tools/pre-execute`, `execute`, `post-execute` waterfall [14]. Standard mode gives the model a file editor, a shell, search, skills, and subagents; Code mode lets the model orchestrate tool rounds in TypeScript; Minimal mode keeps bash and `str_replace_editor`; Creator mode inspects the runtime to author presets [47]. Experimental Agent Teams add a roster, task board, and mailbox over continuable subagents. This is an excellent composition kernel for a *single* agent product. It is not a transaction kernel. Filesystem and subprocess providers share one execution world, so pointing them at a remote sandbox moves Bash and PTY together—and the model still writes files. A durable session event is not a Candidate.

G. Meta-Harnesses

Omnigent is an open-source meta-harness, with a Databricks-operated deployment, that wraps Claude Code, Codex, Cursor, OpenCode, Hermes, Pi, Antigravity, and custom YAML agents [7], [15], [49]. A runner exposes a uniform session API. A server stores policies and sharing. Native Linux wrappers use bubblewrap; macOS uses seatbelt; an L7 egress proxy can attach credentials so the sandbox never holds the real token [15], [16]. That credential-proxy design is one of the strongest OS-isolation stories in this survey. Native terminal wrappers still use tmux. The wrapped harness still performs Git. A policy can deny, transform, or pause a tool call; it cannot mint a Candidate or invalidate evidence when the branch moves.

IV. RECURRING FAILURE CLASSES

The issue corpus and the Git documentation converge on a small set of missing *bindings* [13], [50], [51]. A binding is a rule that makes a class of silent success impossible. Role names, dashboards, and sandboxes do not substitute. Table II summarizes how the surveyed systems bind—or fail to bind—write, completion, evidence, integration, and liveness.

A. Shared Writable Identity

Git worktrees share one object store, one set of refs, and—unless extra machinery exists—one index and one hook namespace [50]. Gas City issue 1181 records implicit pool agents resolving every session to the rig root: two parallel slings clobber uncommitted edits [8]. Gas Town issue 4094 records background polecat operations running `git checkout` against the town-root `.git`, deleting untracked operational files such as `mayor/rigs.json` and Dolt manifests [10]. A worktree is a checkout convenience. It is not a private repository.

`git clone -reference without -dissociate` writes `objects/info/alternates` into a borrowed store. A later `git gc -prune=now` on the mirror can unlink objects a live workspace still names [20], [50]. CoW reflinks `without` a documented fallback have the same ownership question.

B. Hostile Configuration and Inherited Credentials

A repository-local or inherited Git config can install filters, aliases, include files, URL rewrites, credential helpers, hooks, and `protocol.file` transports that execute host code or exfiltrate bytes during an ordinary checkout or diff [50], [52]. Case-folded path collisions on a case-insensitive volume, alternate data streams, and symlink traversal are the same class: Git will write something the operator did not grant. Model-authored `git` in a process that inherited `GIT_*`, `SSH_AUTH_SOCK`, or a forge token is a credential and integrity failure, not a convenience. Compile-time code—Rust `build.rs`, `proc-macros`, TypeScript transformers—executes before any test “runs” [20]. Sandboxing only the test process is insufficient. Detached HEAD, an active sequencer (`MERGE_HEAD`, `REBASE_HEAD`), and a dirty index shared across worktrees convert ordinary “just commit” steps into lost or mis-attributed work [50].

C. Control Plane in Git

Git is a poor coordination database for renewable leases, heartbeats, quota reservations, and unsettled remote effects [51], [53]. Beads-on-Dolt-in-Git is an inventive store and an operationally heavy one: install depends on Dolt, ICU4C, and `bd` [5], and the timeout-after-commit class appears under load [9]. Putting the live ledger in the same bytes the agents edit couples control-plane availability to repository mutation.

TABLE II

AUTHORITY BINDINGS. “TYPICAL HARNESS” IS THE CLAUDE CODE / CODEX / CURSOR / ANTIGRAVITY SINGLE-AGENT PATTERN [18]. BULLET FARM ROWS ARE SPLIT SO A DOCUMENT IS NEVER PROMOTED INTO A DAEMON.

System	Who writes the repo		Isolation	Completion signal	Evidence subject	Integration authority	Liveness oracle
Bullet Farm (designed) [DESIGN]	bullet-gitd only		Private clone, no worktree	Candidate + independent Evidence + forge receipt + observation	base, head, tree, patch, env	Jeryu/GitHub required checks; attestor $\neq$ broker	Lease/fence; ALIVE, DEAD, UNKNOWN, CONTRADICTION
Bullet (implemented) [COMPONENT]	Farm	Production refused	write	Dissociate clone code exists; not reachable online	PENDING then UNKNOWN or FAILED only	One fixture E2 gate; no live subject	Local bare-forge component; Jeryu quarantined
Gas Town v1.2.1	Agent in hook worktree		Git worktree sharing .git	gt done, convoy, Refinery merge	Gates exist; not exact-OID bound	Refinery / local merge queue	tmux session
Gas City v1.4.1	Agent; implicit pool uses rig root [8]		Pack-optional worktree	Formula/run/session closure	Session and bead state	Configurable; still orchestrator-owned	Session + controller
DeepSeek Harness rc.2	Model via fs/shell plugins		Sandbox plugin back-end	Turn/session log	Session events (model-visible)	n/a (not a forge OS)	Agent loop
Omnigent v0.10.0	Wrapped harness		OS sandbox + L7 proxy	Session / policy decision	Policy and session events	n/a (meta-harness)	tmux on native wrappers
Typical harness	single-agent	The agent in the user checkout	Vendor sandbox flags (optional)	Model “done”, PR, or exit 0	Transcript or CI on a branch	Hosted PR or human merge	Process / IDE session

D. Inferred Completion and Unsafe Liveness

Gas Town’s Problems view names the state “Zombie: Dead tmux session” [5]; the multiplexer is the liveness oracle. The corpus shows both failure directions. Issue 3584 reports polecat sessions dying mid-task with no notification while the work stays hooked [11]. Issue 4737 reports a patrol cleanup step whose liveness check classifies every dog as dead and would force-remove live worktrees, held back only by an accident of layout [12]. Issue 4091 reports the refinery aliveness check trusting stale pid files so dead refineries are never respawned [54]. A timeout after Dolt committed server-side is classified transient and retried, yielding up to three complete workflows from one `gc sling` [9]. Hosted agents treat PR-opened as done. Plugin kernels treat turn-end as done. In every case, transport success or process presence is painted as domain truth.

E. Evidence Not Bound to an Exact Subject

A green CI result that names a branch, a worktree path, or a session ID is not evidence about a rebase, a repair commit, or a merge-group composition. Ordinary Git commits have no stable Change identity across rewrite [50], [55]. After the branch moves, yesterday’s green is a different subject. Merge queues that do not invalidate overlapping evidence create silent stale-green admission; queues that invalidate every queued candidate recreate a livelock when arrival rate exceeds verification time [20].

F. Destruction Without Preservation

Cleanup, GC, polecat reuse, and “nuke” operations that target a name, path, or inferred age destroy the only copy of uncommitted work [10], [12], [13]. Preservation must bind an exact incarnation and a sealed artifact *before* deletion eligibility.

V. BULLET FARM ARCHITECTURE

Bullet Farm is a split family: a public hub (`bullet-farm`), a control-plane kernel, a Git capability

daemon, and a projection-only portal [17]. Jeryu is the source forge and the mandatory local effect adapter. GitHub is a separately configured effect adapter, never source authority. Unless a sentence carries another tag, this section is [DESIGN].

A. The Software-Change Transaction

Mission $\rightarrow$ GraphRevision $\rightarrow$ Variant $\rightarrow$ Attempt
 $\rightarrow$ Proposal $\rightarrow$ Candidate $\rightarrow$ Evidence (1)
 $\rightarrow$ Effect $\rightarrow$ Integration $\rightarrow$ Observation.

A Mission is a durable objective plus constraints. A Graph Revision is an immutable package decomposition. A Variant is one hypothesis with an immutable role, routing, context, and configuration snapshot. An Attempt is one execution incarnation under a unique lease and a never-reused fence. A Proposal is typed patch data. A Candidate is a provenance-bound immutable phenotype created only by BulletGit. Evidence is an independently reproduced result over that exact Candidate and admitted gate definitions. An Effect is a brokered remote mutation with read-back. Integration is protected-ref admission by the forge. Observation is post-merge survival. None of those words is a prompt label. Each is a typed subject with an exact identifier [17], [56].

B. Five Trust Planes

ADR 0003 separates control, execution/repository write, independent verification, effect/attestation/integration, and observation/audit [57]. Compromise of the portal, a provider, or a broker must not manufacture independent evidence once those principals exist as distinct identities. ADR 0003 *requires* distinct OS identities on one host and SPIFFE-compatible identities later; that separation is not a current receipt. Production processes still share a UID, and same-UID path-based Git races remain an open setup boundary [23]. The evolutionary control document defines roles as capability envelopes—Planner, Implementer, Critic, Verifier, Effect broker—not as personas that carry credentials [56]. Online learning, larger

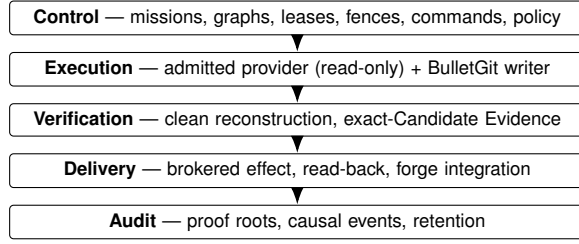


Fig. 1. Five trust planes (design). A plane may consume only grants addressed to its principal. Portal and any evolutionary optimizer are projections or requestors, never mutation credentials [57].

councils, and distributed execution are Phase 9–10 and are out of V1 [56].

C. Three Sovereign Truths

A common instinct is to put missions, leases, and evidence into Git because the product is about Git. That collapses two badly matched stores [51]. Bullet Farm splits canonical truth:

- **Kernel ledger** (SQLite WAL today; PostgreSQL is a refused scaffold [58]) owns mutable authority: leases, fences, commands, policy snapshots, outbox, event log.
- **BulletGit** owns immutable engineering subjects: Changes, Candidates, checkpoints, journals, proof roots.
- **Forge** (Jeryu, optionally GitHub) owns remote refs, required checks, and the merged target SHA.

Cross-boundary values come from generated bullet-wire contracts: RFC 8785 canonical JSON, domain-separated BLAKE3, full 256-bit IDs, algorithm-tagged Git OIDs, and PASETO v4.public envelopes [24], [59]–[61] [COMPONENT]. Prose and TypeScript types do not define the wire.

D. Providers Propose

ADR 0001 is the execution-mode decision [18]. Every provider turn is read-only inside a private workspace generation. The only write input is a validated `PatchProposal`: schema identity, producing `Attempt`, base checkpoint, per-path preimages, and policy-admitted `gate_ids`. Provider text such as `tests_to_run` or a shell command is never argv. Runner resolves gate IDs through a trusted catalog [COMPONENT]. Zero tests, timeout, skipped, flaky, unsupported, and UNKNOWN never equal PASS. At most two bounded repair turns may see typed gate results.

Public mutations use authenticated POST /v1/commands, initially return 202 PENDING, and reconcile through PENDING|APPLIED|VERIFIED|FAILED|UNKNOWN [24] [COMPONENT]. Transport success never implies verification. The Portal renders only those phases and treats watermark mismatch, sequence gaps, and timeouts as UNKNOWN or STALE, never as an empty success list [62] [COMPONENT].

E. Why the Bindings Are the Product

Other systems can copy a role name in a week. They cannot copy a binding without changing what an agent is allowed to

be. A Mayor that still infers liveness from `tmux` is a persona. A fence that is checked inside the ledger transaction at the effects gateway is a binding [63]. A plugin that swaps the filesystem provider is composition. A daemon that is the only process allowed to apply a scoped patch is a binding. A contextual policy that pauses a tool call is governance. An Evidence record that becomes invalid when `head` moves is a binding.

That is the source of leverage. As models improve, a harness that lets the model write Git gets more destructive more quickly. A transaction processor that lets the model propose gets more useful more quickly, because the same fence, Candidate, and verifier contracts apply to a weaker economy model and to a frontier council. The evolutionary document states the same rule for search: hard constraints are filters, never terms a high score may offset, and the optimizer cannot see or learn away the safety envelope [56]. V1 records routing provenance and does not learn online.

F. A Worked #5511-Class Fault

Issue 5511 is the canonical timeout-after-commit race [9], [20]. Under load, `bd create -graph` is killed at the runner deadline after Dolt has already committed. The client sees a timeout, classifies it transient, and re-applies. The result is up to three live workflows and three writers in one checkout.

Bullet Farm’s designed handling is mechanical:

- 1) Graph materialization is one serializable insert keyed by content hash. A replay returns the existing graph; it does not mint a sibling [DESIGN].
- 2) The command that requested materialization is recorded with an idempotency key [64], [65]. An ambiguous timeout becomes UNKNOWN, not a retry [COMPONENT].
- 3) A resolver may adopt a remote or ledger subject only when its identity equals this Attempt’s desired state. Otherwise the subject is ORPHANED and quarantined [DESIGN] (modelled in `EffectCheck`, Section VII).
- 4) Even if two Attempts exist, each requires its own never-reused fence and its own private clone. They cannot share a `cwd` or a `.git` [COMPONENT] (lease/fence and private clone are components; their composition is not).
- 5) A writer that outlives its lease cannot push, comment, or settle: the fence is presented at the effects gateway and rejected inside the ledger transaction [DESIGN].

Items 2 and 4 have component proofs. Items 1, 3, and 5 are specified and are not a live path at HEAD. The point of the walkthrough is not to claim the incident is closed in production. It is to show that the missing primitive is identity-exact adoption under ambiguity, not a smarter Mayor prompt.

G. Evidence Tiers

Writer output cannot satisfy an independent tier [18], [56]. The designed ladder is:

- E0 — untrusted provider or author claim;
- E1 — writer-visible gates on the author’s workspace;
- E2 — clean independent reconstruction of the exact Candidate;
- E3 — hidden or historical holdout, plus hostile suites the author cannot weaken;

- E4 — human review.

The implemented verifier produces typed E2-shaped results for one fixture gate over exact base/head/tree subjects [24] [COMPONENT]. Multi-gate aggregation, executable-digest admission, and E3 holdout custody remain open [DESIGN]. That is the correct partial result: the identity rule exists before the gauntlet is populated.

VI. THREAT MODEL

A. Adversary and Scope

We assume a single-user, loopback V1 host [24]. The adversary is not a nation-state on the wire. The adversary is (i) a frontier model that will try to run shell, Git, and forge tools; (ii) a second concurrent Attempt that shares a name, path, or session identity; (iii) a crashed or partitioned runner that retries after an ambiguous timeout; (iv) hostile repository content (hooks, filters, `build.rs`, `proc-macros`, `.gitattributes`) and repository-resident instruction files the provider will read before any proposal exists [66]; (v) a compromised or confused portal that wants to paint UNKNOWN green; and (vi) an operator who would rather flip an environment variable than mint a lease-bound grant.

Out of scope for V1, and therefore not silently promised: multi-tenant SaaS, remote untrusted runners, cross-repository sagas [67], online routing learners, a compromised Kernel ledger or `bullet-gitd` process (either is total loss on a single-user host), and a formal proof that physical side effects are exactly-once. Those absences are named so a red team cannot treat them as implied.

B. Trusted Computing Base (Design)

The designed trusted computing base for an admitted write is the Kernel ledger transaction, the signed launch-grant verifier, the egress-isolation receipt, `bullet-gitd`'s final-authority checker, and the forge's required checks [DESIGN]. The model, the Portal, MCP servers, vendor sandboxes, and `tmux` are untrusted. Confused-deputy discipline is why a portal, an MCP server, or a vendor sandbox must never inherit forge credentials [68]. Today no admitted write exists, so the TCB is a specification of what a future [TRANSACTION] receipt must exercise.

C. Implemented Boundaries

Each item below is [COMPONENT]: a named commit and mapped tests exist (Appendix B). None composes into a transaction.

- **Fail-closed argv gate.** The supervised `argv` boundary enforces a kill switch, a `worktree/tmux` deny list, the exact admitted executable, and default refusal of live provider programs; supervision kills the POSIX process group on exit, crash, cancellation, heartbeat loss, or deadline [58].
- **Provider process admission.** One absolute canonical executable is compared to its BLAKE3 digest and a fresh runtime probe; a unique mode-0700 HOME is created; only exact policy-listed OAuth files are copied, after digest and symlink checks, as mode 0400; the child environment is built from locale hints plus `HOME/TMPDIR/XDG`

paths, so host `PATH`, `SCM`, `cloud`, `SSH`, and `API` secrets are not inherited; canary scanning covers environment, `stdout`, `stderr`, normalized events, and the validated proposal [58], [69].

- **Signed launch grants.** A PASETO v4.public token binds the durable active lease (Mission, repository, graph revision, work package, Variant, Attempt, fence, runner epoch, workspace nonce), the evaluated admission (path, digest, profile, protocol), the policy snapshot digest, a budget reservation, and a single-use nonce; audience `provider-runner`, operation `launch-provider`, $TTL \leq 15s$; minted in-process from an operator-held mode-0600 key with no HTTP route; the verifier clears only `SIGNED_ADMISSION_UNAVAILABLE` [61], [70].
- **Linux egress isolation.** The provider runs in a fresh user+network namespace with an `nftables` default-drop ruleset permitting only loopback and TCP to a host-side allow-listing `CONNECT` proxy; eight in-namespace probes (direct internet, host forge port, host decoy port, DNS over TCP and UDP, proxy reachable, proxy-disallowed host, proxy-allowed path) must produce exactly their expected refusals and admissions before the child starts; the sealed receipt clears only `EGRESS_ISOLATION_UNAVAILABLE` [70]. This was exercised on one development host (live lane 3/3); it is containment-component evidence, not provider conformance.
- **Policy refusal.** Under the committed generation-1 `v1alpha1` snapshot [71], a fully valid grant with both blockers cleared still ends in `POLICY_LIVE_ADMISSION_DISABLED` without spending its nonce [70], [72].
- **Runner self-fence.** The runner keeps a local monotonic self-kill deadline at $now + \frac{4}{5}TTL$, strictly shorter than the server expiry, and freezes on the deadline or on a zero-row renewal; this is the STONITH direction of the fence-mediated liveness exit [58], [73].
- **BulletGit authority gateway.** In a fresh daemon, `clone` reaches the unavailable final-authority checker and returns `AUTHORITY_CONTRACT_UNAVAILABLE` before any repository I/O; later verbs fail `NOT_CLONED`; the unsigned legacy token grants nothing [52]. Hostile-git refusal, dissociate cloning under a mirror lock, scope and collision refusal, and preservation-before-cleanup are components (Section IX).
- **Portal honesty.** One-time bootstrap to an `HttpOnly/SameSite` cookie plus `CSRF` value; exact HTTP 202 with a validated `PENDING` subject; every status read repeats the admitted command id; timeout and sequence gaps render UNKNOWN or STALE; eleven of fifteen catalog surfaces render explicit unknown because no `farmd` projection exists [62].

D. Boundaries That Are Not Implemented

Each item below is [DESIGN] or [BLOCKED]. Naming them is the point.

- **Signed internal lease transport.** The existing `HttpLeaseClient` is unauthenticated, self-asserts

scheduling and runner identity, and its release/advance are not full-subject atomic operations; it must not be mounted. A signed contract binding runner, authority epoch, request digest, and the full lease subject is a predecessor of any transaction [24], [70].

- **S1/S2 containment.** ADR 0007’s rootless-OCI S1 (namespaces, cgroup v2, seccomp, read-only root, deny-by-default network) and separate-kernel S2 backends are designed; Landlock, seccomp, and namespaces are the intended Linux primitives [69], [74]–[76]. The egress component above covers the network dimension only.
- **Online mutation permits and settlement.** One-second PASETO mutation permits, Kernel online reservation and final check, and typed settlement are specified [61]; BulletGit records the shape locally and refuses to proceed without them.
- **Hidden holdouts (E3) and oracle-modifying diff classes.** Required for R2+ by C2; no holdout store or custodian exists [21], [77].
- **Attestor $\neq$ broker.** The check-publishing path that re-computes a verdict from a signed proof bundle with its own credential (C4) is designed; live forges are blocked [78].
- **Distinct OS identities per plane.** Required by ADR 0003; processes share a UID today.
- **Kernel v1alpha2 loader.** The hub validator accepts a generation- ≥ 2 live-admission policy with a registered provider-runner key; the Kernel loader still accepts v1alpha1 only [72].
- **Content-addressed artifact storage, restore admission, and audit anchoring** [58], [77].
- **Same-UID pathname swap** between tool verification and Cargo/Node/npm exec during source setup is an accepted residual until a signed prebuilt installer exists [23].

VII. FORMAL MODELS: WHAT THE TWO TLC MODELS ENCODE

Exactly two protocols are model-checked in TLA+ [79], [80] and pinned by a model lock [81] [COMPONENT]. The pin fixes the TLC release (1.7.4, tool version 2.19) by SHA-256 and SHA-1, requires Java 21, refuses a third module, and fails on any drift of module bytes, configuration bytes, or single-worker state counts. This section states what the models check so that no reader infers more.

LeaseFence models two runners, one lease, a monotonically increasing fence, a database clock, scope revisions with a mutation barrier, freeze, restore epochs, and apply attempts. Constants: `Runners = {r1, r2}`, `MaxFence = 3`, `MaxTime = 4`, `MaxScope = 3`, `MaxOps = 2`, `MaxEpoch = 2`, `MaxFreeze = 3`. The checked invariants are exactly `TypeOK`, `BarrierHasNoInFlightApply` (a scope barrier never coexists with an in-flight apply), and `RestoredAuthorityNeedsFreshAcquire` (no in-flight apply while no lease is held). Stale-fence, stale-epoch, expired, frozen, barrier, and unacknowledged-scope applies are refused by the `Authorized` guard and counted, not asserted as a separate invariant. Fence monotonicity holds

by construction (the fence changes only in `Acquire`, by `+1`); it is not a checked property. Pinned run: 529,851 states generated, 128,913 distinct, depth 24.

EffectCheck models one effect and one dependent check through the phases `absent`, `intent`, `dispatching`, `unknown`, `verified`, `orphaned`, against a remote that may be untouched, desired, or mutated by a third party. Constants: `MaxDispatches = 2`, `MaxEpoch = 2`. The checked invariants are exactly `TypeOK`, `AtMostOneLogicalEffect`, `VerifiedEffectWasReadBack`, `CheckRequiresDurableProof`, and `ThirdPartyStateIsNeverAdopted`. A lost response leaves the effect in `unknown`; read-back adopts it only when the remote holds the desired state, marks it `orphaned` when a third party wrote, and returns to `intent` when nothing was written. Pinned run: 1,585 states generated, 378 distinct, depth 18.

What the models do *not* do, stated so the limitation is citable:

- Deadlock checking is disabled in both configurations (`CHECK_DEADLOCK FALSE`); no liveness or fairness property is checked. The models are safety-only and bounded.
- Wound-wait, multi-resource scope expansion, and two-track amendment (A1, C3) are not modelled. The historical adjudication’s own criticism—that a lock model showing only the no-conflict path proves nothing about deadlock freedom—still applies to the family’s models, which make no such claim [20].
- `AtMostOneLogicalEffect` is enforced by assignment: the logical-effect counter is set to one rather than incremented on each remote mutation, so the invariant cannot fail by construction. The real protection against a second physical write is the guard that dispatches only while the remote is untouched. A stronger model would count physical writes.
- `EffectCheck` defines `NoDispatchAfterStop` (no dispatch while policy is expired or the system is frozen), but the configuration does not list it, so it is not checked.
- Neither model represents Git, a forge, a provider, two Attempts on one Variant, or any cross-plane composition. They prove two protocols under small bounds. They do not prove the product [21].

The three JSON traces under `formal/traces/` are executable conformance fixtures replayed by Rust tests (`formal_traces, 2/2`); they are not model-check results [81].

VIII. RED-TEAM DISPOSITION (C1–C12)

The family `TEAM.md` document is a hostile review of the Centerrail specification [21]. It is historical provenance, not runtime authority. Living control is the evolutionary document, the release contract, and the product-gap register [23], [56], [82]. We adopt the twelve critiques that simplify or harden with no regression, and we keep the four rejected cuts rejected: the verifier plane stays; SQLite remains the V1 ledger; the five planes stay; the forge remains merge-sovereign.

Table III is the product status of those twelve hardenings. “Adopted” means the rule is the current design. “Component” means a named commit and mapped tests exist. “Designed” means the rule is specified and not a live path. A designed row is not a silent implication that the tree enforces it.

C7 is the one place the original build plan over-reached and the tree already followed the fix: exactly two TLA+ models, not seven, before Phase A code. C8 was adopted by the historical review and then superseded: the frozen V1 closure plan requires conformance receipts for all four adapters [18], [24]; this paper reports the frozen rule. C2 and C9 remain the load-bearing unshipped hardenings: without an oracle the author cannot weaken, and without identity-exact adoption, a timeout-after-commit still launders a stale write. Those two are why a connected TRANSACTION_PROOF is the next product, not a larger council [82].

IX. BULLETGIT: THE MISSING SUBSTRATE

Models and harnesses commoditize. An agent-first source-control substrate that understands authority, change identity, proof, lineage, and protected integration does not [51]. BulletGit is that substrate. `bullet-gitd` is the sole writer of the private workspace [52]. Pack protocol, refs, and protected updates remain ordinary Git at the forge. Unless tagged otherwise, the mechanisms in this section are [COMPONENT]: they exist in `bullet-git-workspace` and `bullet-gitd` with mapped tests, and none is reachable through a production clone.

A. Private Clone, Not a Worktree

Workspaces never clone the operator’s source tree directly. A bare mirror is created or fetched under an exclusive lock. The private clone runs

```
git clone -reference-if-able <mirror>
        -dissociate
```

while the lock is held [52]. Objects may be borrowed during the copy; no alternates file survives. Mirror GC cannot corrupt a live workspace. Immediately after clone, `origin` is removed, `credential.helper` is emptied, `GIT_ASKPASS` points at a deny script, `GIT_TERMINAL_PROMPT=0`, and `GIT_SSH_COMMAND=false`. A model-issued push has no destination and no authenticator.

A `.git file`—the on-disk shape of a worktree—is `WORKTREE_FORBIDDEN`. Sequencer files at checkpoint time are `SEQUENCER_ACTIVE`. Checkpoints run `git write-tree` against a temporary `GIT_INDEX_FILE`, never the live index.

B. Hostile-Git Refusal

The child environment is cleared of inherited `GIT_*` variables and rebuilt with per-workspace `HOME/XDG`, `GIT_CONFIG_NOSYSTEM=1`, and `GIT_CONFIG_GLOBAL=/dev/null` [52]. Every invocation disables paging, hooks, credentials, filesystem

monitors, external attribute files, and signing, and passes `protocol.file.allow=never` except for the one local clone that needs file transport. Before a repository-scoped command, BulletGit parses local config with includes disabled. Command-bearing or truth-redirecting filter, alias, include, pager, URL rewrite, remote helper, diff/textconv, merge driver, submodule, credential, signing, editor, sparse, SSH, hook, worktree, or external attribute settings fail closed as `HOSTILE_GIT_CONFIG`. A real-repository test plants a clean filter and a canary behind `.gitattributes` and proves refusal before execution.

C. Scope, Identity, Preservation

`apply_change` validates every path against a `ScopeGrant` before writing: Unicode NFC, no `...`, no `.git` component, no backslashes or NUL, no alternate data streams, no trailing dot or space, no symlink traversal or symlink targets. Duplicate and case-folded collisions refuse the whole batch; batches above 1,024 operations or 1 MiB per body refuse. `CandidateId` is content-derived from change, tree, and head with length-framed hashing, so two trees under one Change cannot share an identifier; `patch_hash` is BLAKE3 of the `git diff base...head` bytes [52]. A provenance-complete `Candidate identity` (distinct `ContentId`, complete proof manifests) is still open [DESIGN] [24]. Local cleanup accepts only an opaque, daemon-sealed preservation receipt that re-verifies source, destination, subject, journal, bundle, and full artifact immediately before deleting exactly the sealed work directory; a synced tombstone survives; post-delete ambiguity is typed `UNKNOWN`; online settlement is not a live path [52].

D. Honest Production Boundary

The public daemon constructor installs an unavailable final-authority checker. In a fresh daemon, `clone` returns `AUTHORITY_CONTRACT_UNAVAILABLE` before repository I/O [52]. Later verbs fail `NOT_CLONED`. The unsigned legacy token grants nothing. Production enablement still requires a published immutable `bullet-wire` tag, local PASETO verification, Kernel online reservation, and one-second operation permits [61]. That fail-closed posture is a feature of the current tree, not a footnote.

X. EXACT INVENTORY: WHAT WORKS TODAY

Table IV is a receipt table, not a feature list. Subjects are those recorded by the 2026-08-25 closure reconciliation [23], [24], [82]: `hub 0cc7eecffd4a6fe51c0288ea4d80e02729f79848` (code/report subject under Hub reconciliation head `6eb002b`), `kernel 0346fd531e231a88e5b1eb3a8257be04f235068b`, `BulletGit 236f4efe2b8243b3c4dd5da13269e45119b061df`, and `portal 8272844e44d89ea3b26a5b3c4f25909a069e101`. Those commits include the operator-ratifiable `v1alpha2` validator, the synchronized schema-bundle consumers, sealed setup-tool subjects, and the

TABLE III

CENTERRAIL C1–C12 DISPOSITION AGAINST THE INVENTORY HEADS. ADOPTED IS DESIGN LAW. COMPONENT IS A RECEIPT. DESIGNED IS SPECIFIED AND NOT SHIPPED.

ID	Adopted meaning	Product status
C1	Every invariant is T1 schema / T2 gateway / T3 test	Registry exists and validates; three rows enforced at Gate 0 (C1, C7, canonical/registry); all other C/EV rows planned [83]
C2	Oracle-modifying diffs; holdouts required for R2+	Designed; implemented verifier is one fixture E2
C3	Two-track scope expansion [84]	Accepted ADR; not a live Attempt path
C4	Attestor $\neq$ broker; reconstructible check	Designed; live forge blocked
C5	CONTRADICTORY has fence-mediated exits	Designed; runner self-fence is a component
C6	Bounded probe reservation; unknown $\neq$ headroom	Designed; unknown_quota_is_headroom=false is a policy invariant
C6b	One seat-equivalent per named human	Designed; not enforced in the kernel
C7	Formal-model exactly two Phase-0 protocols	Adopted; model-lock receipt (Section VII)
C8	GA = kernel + any two certified providers	Superseded for this V1: the closure plan requires all four adapters; live still operator-gated [18]
C9	Identity-exact adoption (fence + desired OID)	Command idempotency component; EffectCheck models the rule; graph mint not live
C10	Verifier dwell back-pressures writers	Designed
C11	Freeze chip: recorded vs enforced-on-N/M	Portal honesty component; countdown designed
C12	Saga quarantines blast radius, not the fleet	Out of V1

deterministic release-truth projection. The portal bundle root at that subject is

```
blake3:556d91f8504299230bc75e027108d1d943fd9ff9
9303cea7c2a155ee69edd973
```

A later commit invalidates a row until the mapped gate is replayed. Earlier preprint drafts named hub 74507b3 / kernel ba70d8b / BulletGit 5dac98e / portal d607068; those are superseded.

Kernel architecture is equally explicit about what the bins are not [58]. bullet-farmd records PENDING and an internal reconciler can settle known demo work only to UNKNOWN or unsupported kinds to FAILED. It cannot emit APPLIED or VERIFIED. Dispatch remains policy-disabled. Hub commit bf5c642 accepts a vlalpha2 snapshot only when policy_generation ≥ 2 and an unrevoked provider-runner PASETO key is registered; the Kernel policy loader still accepts vlalpha1 only, so a vlalpha2 file is not an admission path [23], [72]. The four provider crates remain transcript machines until that operator ratification and a loader mirror exist. Feature-gated tests that pass by refusing spawn are quarantine evidence, not conformance [25]. Unauthenticated farmd lease HTTP is not an admission path and must not be remounted to mint a grant [70]. Clearing SIGNED_ADMISSION_UNAVAILABLE does not clear EGRESS_ISOLATION_UNAVAILABLE; those are independent blockers. That stack is the opposite of an environment variable: a shared BULLET_LIVE_ADMISSION token was rejected as admission and removed [70].

A. The Archived Live Demonstration

One archived receipt from 2026-08-24 14:04Z exists in the family evidence bundle [85] [ARCHIVED]. It records, on real processes on one host: one Mission materialized exactly once; fences 1 then 2 with the stale fence refused; a Candidate with real Git OIDs (base 859fc6cf, head e4701435, tree 336a45c4) whose actual scope was one file; a writer gate that passed with zero repair rounds; an E2 Evidence result produced by the separate bullet-verifier clean-room binary; a local effect on a bare forge component recorded COMMITTED with read-back verified; and Jerry

LIVE_FORGE_QUARANTINED with no authentication. Two planners (Claude, Codex), a fusion step (Cursor, FUSED, not degraded), and a runner turn (Claude) ran as real provider sessions; the metered Claude cost recorded in the receipt is 0.62 USD and the four provider turns took about 74 s of wall clock in total.

The receipt classifies itself LIVE_DEMONSTRATION with transaction_gate_eligible=false. It was produced on kernel commit a9a6ae2 behind an environment-token admission gate that ADR 0011 later rejected as not being admission; that surface is not in the tree at the inventory HEADs, and HEAD policy keeps live_admission_enabled=false. The receipt therefore proves that the seven-clause transaction can run once end-to-end on real processes under a mechanism the family has since refused. It certifies nothing about the current tree and clears no release gate [23].

XI. EVOLUTIONARY CONTROL AS DESIGN

Everything in this section is [DESIGN]. The evolutionary document is normative for what “evolutionary” may mean in V1 and states explicitly that it is not a claim that any runtime property is complete [56]. The committed policy pins evolutionary_authority=false [71], the closure plan gates the program on a real single-lane transaction, and the product-gap register lists it as G11, explicitly post-V1 [82].

A. What Evolves and What Cannot

The unit of evolution is an immutable, content-addressed TeamRecipe, not a live agent. A recipe may name typed roles and their input/output contracts, communication edges and context-capsule policy, certified provider/model/profile choices, bounded concurrency, token, cost, time, repair, and verifier budgets, and deterministic stopping, escalation, dissent, fusion, and fallback rules. It cannot contain or evolve credentials, authority claims, safety rules, risk classes, protected paths, evidence floors, hidden evaluators, integration policy, effect permission, or hard budget ceilings. A new recipe creates new Attempts and workspaces; it never inherits a live

TABLE IV
SELECTED COMPONENT RECEIPTS VERSUS PRODUCT GATES
(2026-08-25). EVERY COMPLETE ROW IS COMPONENT_PROOF: A
NAMED COMMIT AND MAPPED TESTS EXIST. NO ROW IS A RELEASE.

Subject	Class	Boundary
Signed wire DTOs, JCS, PASETO shapes, launch-grant contract (hub; 83 tests)	[COMPONENT]	No running issuer; no immutable published tag
DB-clock leases, TTL 1..15 s, never-reused fence; runner self-fence Atomic	[COMPONENT]	No signed final-authority mint
lease/command/event/outbox; authenticated 202 ingress; UNKNOWN/FAILED worker	[COMPONENT]	No APPLIED/VERIFIED path; public dispatch of providers/effects open
Four offline provider parsers (Claude, Codex, Cursor, Antigraity); strict recursive JSON	[COMPONENT]	Public spawn refused; no live conformance
Provider admission: digest-bound binary, 0700 HOME, 0400 OAuth copies, positive environment, canary scan	[COMPONENT]	Dispatch blocked without signed authority and egress
Signed launch grant + Linux egress isolation (eight probes)	[COMPONENT]	Live dispatch policy-disabled; no HTTP mint [70]
Policy v1alpha2 validator (hub)	[COMPONENT]	ADR 0012 Proposed; Kernel loader v1alpha1-only; no committed policy enables live admission [72]
Admitted fixture gate + bounded verifier transport	[COMPONENT]	One fixture gate; no JSON-RPC production path
Fail-closed bullet-gitd + dissociate clone + hostile-git + preservation + tombstone	[COMPONENT]	Production clone refused
Portal CSRF/202 + PENDING→UNKNOWN + SSE STALE; exact bundle root	[COMPONENT]	Not packaged/embedded; 11 of 15 catalog surfaces stay explicit UNKNOWN
Setup refuses schema-2 hub-only install; signed bundle verify/extract	[COMPONENT]	No signed schema-3 lock or prebuilt installer
Two TLC models with pinned state counts	[COMPONENT]	Safety-only, bounded; deadlock checking off; no wound-wait (Section VII)
Offline transaction demo (just demo)	[SYNTHETIC]	Simulator boundaries; not gate-eligible
Live demonstration receipt, 2026-08-24 14:04Z	[ARCHIVED]	Outside the ladder; not reproducible at HEAD
Five-plane TRANSACTION_PROOF	[BLOCKED]	No signed cross-plane crash receipt at HEAD
Live provider / Jeryu / GitHub effects	[BLOCKED]	Operator ratification and forge credentials
Hub Jankurai ≥ 90, zero hard findings	[BLOCKED]	58 (raw 58), 10 caps, 44 findings: 29 high/hard and 15 medium/soft, at the 2026-08-25 audit [23]
Family check release	[BLOCKED]	26/26 gates; static negative inventory
Evolutionary recipe archive	[DESIGN]	Gated on a working transaction; not started

session, credential, workspace, proof, or privileged cache from a parent [56].

Hard constraints are filters, never terms that a high score may offset. Campaign admission first applies a deterministic feasibility shield—authority, containment, provider/profile certification, role conflicts, quota, budget, verifier capacity, task-policy compatibility—and infeasible recipes consume no

provider or verifier budget. Every task class retains a strong single-agent recipe as the incumbent and fallback; a multi-agent recipe earns eligibility only when matched evaluation shows that its surviving verified outcomes justify coordination, provider, and verifier cost. Agent count, transcript volume, confidence, and activity are never fitness dimensions [56].

B. Archive, Ladder, Rollback

Eligible results enter a bounded multi-objective quality-diversity archive in the MAP-Elites family [86], [87]. Human-versioned behavioral descriptors may distinguish niches such as task/risk class, team topology, provider diversity, latency band, cost band, and verifier demand; descriptors must be observable from receipts and must not encode hidden-answer proxies. Within each cell, hard constraints filter first; the frozen policy then retains a Pareto set over independently measured correctness, robustness, cost, latency, intervention, and observation survival, with a stable content-ID tie-break. No global scalar reward can trade a safety regression for throughput. Archive cells, descriptors, objectives, and bounds are versioned policy subjects; changing one starts a new campaign rather than rewriting history.

Recipes advance only through comparable completed rungs, in the successive-halving spirit of staged budgets [88]–[90]: B0 deterministic simulator, property, contract, and fault checks; B1 a small visible matched task block against the incumbent; B2 a larger visible block under the same total compute; B3 an untouched sealed confirmation under independent oracle custody; B4 no-effect shadow routing on live task classifications; B5 a bounded low-risk canary with automatic rollback; B6 an eligible routing arm with an incumbent traffic reserve. Pruning happens only between completed, comparable rungs. Raw holdout failures never enter author context, and a recipe family that authored a Candidate cannot provide its independence-required review [56].

Conservative routing chooses only among already certified recipes, records the eligible set and abstentions, and preserves a fixed incumbent reserve. One champion may coexist with multiple niche challengers. A sparse island model [91] may copy only immutable recipe identity between campaigns, and the destination must re-evaluate it independently. Post-promotion monitoring covers escaped defects, revert rate, duplicate or ambiguous effects, false completion, cost, latency, verifier backlog, human intervention, and provider or task-mixture drift; a bound breach removes the recipe from routing, preserves lineage and negative knowledge, and falls back to the certified incumbent. Contextual routing learners [92] are post-V1 and require holdouts, causal or offline evaluation [93], rollback, drift detection, and an explainable signed decision record before influencing production admission. V1 does not update weights, prompts, policies, or privileges online.

C. Adjudicated Hardenings A1–A7

The historical adjudication added seven hardenings to C1–C12 [20]. Table V is their status; two proposed fixes (identifier renaming before blind review, blanket transitive locking of trait implementors) were rejected as regressions and stay rejected.

TABLE V
ADJUDICATED HARDENINGS A1–A7 AGAINST THE INVENTORY HEADS.

ID	Hardening	Status
A1	Wound-wait for multi-resource scope expansion [94]	[DESIGN]; not in LeaseFence
A2	Private clones via <code>-dissociate</code> or CoW reflink	[COMPONENT] dissociate; reflink fast path designed
A3	Formatting normalization and watermark scrub before blind review (no renaming)	[DESIGN]
A4	Staged race budgets (about 15 % per Variant, arbiter culls, survivors continue)	[DESIGN]
A5	Compile = code execution: build scripts and macros run	[DESIGN]; one fixture gate
A6	network-denied in the gate sandbox Tree-disjoint evidence preservation on integration	[DESIGN]
A7	STONITH timing: runner self-kills before server expiry	[COMPONENT] ($\frac{1}{3}$ TTL deadline); grace-vs-TTL policy invariant designed

XII. COMPARISON

Tables II and VI use pinned subjects: Gas Town v1.2.1, Gas City v1.4.1, DeepSeek Harness dsh-v0.1.1-rc.2 (b150a55), Omnigent v0.10.0 (40755dd) [6], [19], [43], [45], [49]. Empty knowledge is written `unknown`, never guessed. Bullet Farm is split into the designed contract and the implemented tree so a reader cannot promote a document into a daemon.

The comparison is deliberately unfair to Bullet Farm on maturity and fair to everyone on bindings. Gas Town is a shipped orchestrator with substantial operational texture. DeepSeek is a composable agent product. Omnigent is a collaboration and governance layer that can wrap the others tomorrow. Bullet Farm is pre-release. What the others do not have—and what makes the architecture categorically different—is the set of bindings in the first row of each table. A system can add a Mayor, a plugin, or a policy function and still retry an `UNKNOWN` push into a shared worktree.

A. What We Must Not Lose From Each System

Gas Town’s factory metaphor is the right product skin for a fleet: named work, visible roles, escalation, and a merge rail an operator can watch [5], [13]. Bullet Farm should keep those surfaces as projections over the event log, not as authority-bearing model personas. Gas City’s primitive test—keep judgment out of transport, prefer primitives that remain useful as models improve—is compatible [19]. The added Bullet rule is that transport success, task closure, session state, and model judgment cannot confer mutation, verification, or integration authority.

DeepSeek’s session-log invariant—model-visible means logged—is the correct epistemology for a *cognitive* plane [14]. Bullet Farm needs the same discipline for prompts, tool schemas, and injected context. It must not use that log as repository truth. A reconstructable transcript proves what the model saw. It does not prove what landed on `main`.

Omnigent’s L7 credential proxy is the correct shape for forge and provider secrets: the sandbox never holds the real token; the proxy attaches it on approved egress [7], [16].

ADR 0007 and the provider-admission crate already strip host secrets and project minimum OAuth copies [58], [69]. The remaining work is audited provider-only egress on every path, not a decision to put `GH_TOKEN` back in the child environment. Omnigent’s multi-device session sharing is a collaboration product. Bullet Farm’s Portal is the opposite product on purpose: a sequence-bound projection with no skip-checks affordance [62].

Typical single-agent harnesses remain the right *proposal producers*. V1 GA requires conformant Claude, Codex, Cursor, and Antigravity adapters under one proposal/evidence contract [18], [24]. Wrapping them (Omnigent) or replacing their loop (DeepSeek) is a different layer. Bullet Farm’s layer is the transaction around them.

B. Interoperability Without Authority Leakage

ACP and MCP are emerging interoperability layers [26], [32]. They should be consumed as capability observations and tool seams, never as mutation authority. A Cursor ACP session that streams a patch is a provider transcript. An MCP server that can run `git` is a hostile tool until it is reduced to a typed proposal or an admitted gate. The same rule applies to vendor “plan” modes: they are useful read-only observations. They do not admit a write.

XIII. IDENTITY, QUOTA, AND THE HUMAN SEAT

This section is [DESIGN]. Gas Town’s scheduler is a rate governor (`scheduler.max_polecats`) aimed at 429s [5]. That is necessary and not sufficient. The Bullet Farm identity design names profiles with owner, authorization class, custody mode, credential generation, and health [20], [56]. Personal CLI subscriptions stay in local custody; API and enterprise credentials use envelope encryption. Login success comes from an adapter `auth_status` probe, never from a UI button. Quota observations are typed and ranked: official API, structured event, headers, typed limit, estimator, then `unknown`. `unknown` is never scheduled as headroom; the committed policy already fixes `unknown_quota_is_headroom=false` [71]. A bounded probe reservation may convert `unknown` to `known` without scheduling bulk work on a guess.

Named human subscriptions are rate-governed to one seat-equivalent per human regardless of how many consumer logins an operator can collect; throughput must come from `api_credit`, `organization_seat`, or `service_identity`. That rule is a compliance-as-code stance, not legal advice, and it is not enforced in the current kernel. It is recorded because it is the subsystem orchestrators most often omit.

XIV. RESOLUTION MAP

Table VII is how this paper “addresses” the failure classes without claiming they are all enforced in production. Status uses the closure vocabulary and the class tags of Section II [24].

The adjudicated hardenings A1–A7 are dispositioned in Table V. The paper adopts the adjudication’s discipline: a fix

TABLE VI
GIT AND WORKSPACE SUBSTRATE. “IMPL.” IS THE BULLET FARM TREE AT THE SECTION X SUBJECTS.

System	Private clone	Dissociate / CoW	Hostile-Git refuse	Path scope grant	Stable Change/Candidate	Preserve-before-destroy
Bullet Farm designed	required	required	required	required	required	required
Bullet Farm impl.	code present; production clone refused	-reference-if- ab implemented -dissociate SafeGit	in implemented	in implemented	wire-shaped IDs; provenance-complete Candidate still open	implemented locally; online settlement open
Gas Town / City	no (worktree / shared cwd)	not the isolation story	unknown as a kernel	optional scripts	pack bead IDs, not Candidate OIDs	reuse/reset; issue 4094 class
DeepSeek Harness	unknown (workspace = fs provider)	n/a	plugin policy, not Git parser	tool/sandbox policy	session IDs	session persist, not sealed generation
Omnigent	sandbox FS, not a private .git	n/a	OS + credential proxy	policy handler	session IDs	sandbox lifetime
Typical harness	user's checkout	rare / foot-gun if used	inherited Git config	prompt or ignore file	Git commit SHA only	leftover branches

TABLE VII
FAILURE CLASS TO BULLET FARM RULE. STATUS IS HONEST AGAINST THE INVENTORY HEADS, NOT AGAINST THE SPECIFICATION.

Failure class	Rule	Status
Shared worktree / shared cwd	Ban worktrees; private clone per Attempt; .git file ⇒ WORKTREE_FORBIDDEN	[COMPONENT]; production clone refused
-reference GC race	-dissociate (or CoW reflink) while mirror lock held	[COMPONENT] dissociate; reflink designed
Hostile Git config / hooks	Clear GIT_*; parse local config; refuse filters/aliases/includes	[COMPONENT]
Model git / inherited tokens	Provider read-only; empty credential helper; no remote; positive environment	[COMPONENT] admission; live spawn blocked
Control plane in Git/Dolt	SQL ledger for leases; BulletGit for Candidates; forge for refs	[COMPONENT] ledger and journal; not one transaction
tmux / silence as liveness	Lease + monotonic fence; four-valued observation; STONITH self-kill	[COMPONENT] lease/fence/self-fence; full observation designed
Timeout-after-commit duplicate mint	Idempotent graph materialize; UNKNOWN never auto-retried	[COMPONENT] command idempotency; graph mint not live
Stale-green after rebase	Evidence names exact OIDs; move invalidates; tree-disjoint reuse is a whitelist	[COMPONENT] fixture E2 only
Compile = code execution	Gates run in the writer/verifier sandbox class, network-denied [69]	[DESIGN]; one fixture gate
Cleanup by name/path	Preserve sealed generation; cleanup only by receipt	[COMPONENT] local preservation
Fleet as merge authority	Forge required checks; attestor ≠ broker	[DESIGN]; live forge blocked
Inferred empty/healthy UI	Watermarked projections; UNKNOWN ≠ empty success [62]	[COMPONENT] on projected surfaces

that destroys reviewability or serializes the repository is not a fix [20].

XV. EVALUATION AND FALSIFICATION

A. What a Fair Benchmark Would Be

A fair comparison uses one repository/task corpus, one tool budget, one wall-clock limit, the same provider profiles, the same retry cap, and the same integration policy [19]. Report distributions, not a single success rate. Minimum metrics: in-window Candidate survival and protected-integration rate; escaped-defect and revert rate; false-completion and zero-test rate; duplicate or ambiguous effect rate; crash-recovery time and lost-work bytes; spend, elapsed time, and human interventions; verifier backlog p50/p95.

No benchmark begins while synthetic proof substitutes for the five-plane transaction. No public comparison is published without raw task subjects, configuration, receipts, exclusions, and the exact upstream commits in Section XII. The first legal corpus is therefore: one Mission that materializes exactly once under kill-9 at every plane; one Variant that takes a never-reused fence; one private clone with no surviving alternates; one independent E2 reconstruction the author cannot weaken; one brokered effect whose timeout stays UNKNOWN until identity-exact read-back; and one portal snapshot that refuses to paint that UNKNOWN green [24], [82]. Until that saga has a signed receipt, Tables II and VI are architectural comparisons, not performance results.

B. What Would Reject the Design

The design is rejected—not “improved in a later version”—if any of the following is observed on an admitted build:

- a stale fence is accepted for a mutation, effect, or settlement;
- one command materializes two live graphs or two writers in one checkout;
- writer-produced proof satisfies an independent Evidence tier;
- UNKNOWN, timeout, zero tests, or a skipped gate is stored or rendered as VERIFIED;
- a worktree, shared cwd, or surviving alternates file is admitted as a writer workspace;
- the portal paints an unwatermarked or contradictory snapshot as a healthy empty list.

Those conditions are executable. They are the opposite of a scoreboard.

C. What Would Falsify This Paper

The design conditions above need an admitted build. The following conditions need only the commits in Appendix A and the commands in Appendix B, and each falsifies a specific sentence of this paper rather than the design:

- 1) Any [COMPONENT] row of Table IV whose mapped test command fails, or is skipped, at the named commit falsifies that row.
- 2) bullet-family check release -json reporting fewer than 26 BLOCKED gates at hub 6eb002b

falsifies Section X; more than 26, or a gate passing, falsifies the sentence that no static placeholder counts as passing evidence.

- 3) `formal/model-check.sh` producing state counts other than 529,851/128,913 and 1,585/378, or accepting a third module, falsifies Section VII.
- 4) A provider binary spawning under the committed `v1alpha1` policy, by any path other than the refused environment token, falsifies the “policy refusal” boundary.
- 5) A fresh `bullet-gitd` accepting `clone` without `AUTHORITY_CONTRACT_UNAVAILABLE` at Bullet-Git 236f4ef falsifies Section IX.
- 6) Any Gas Town or Gas City issue cited in Section IV not describing the failure class attributed to it, at the URL and date in Appendix A, falsifies that attribution. (This revision removed one such citation; see Appendix C, attack 1.)
- 7) The archived receipt’s fields differing from Section X falsifies that subsection; the receipt succeeding on the inventory HEADs would falsify the sentence that the surface is absent, and would be welcome.

XVI. WHAT THIS PAPER DOES NOT CLAIM

A red team that needs a list of forbidden promotions can stop here.

- We do not claim measured superiority on latency, cost, autonomy, safety, or defect escape against Gas Town, Gas City, DeepSeek Harness, Omnigent, or any single-agent harness.
- We do not claim a connected five-plane `TRANSACTION_PROOF` at HEAD, a hub-only installer, or a signed five-target release [82].
- We do not claim live Claude, Codex, Cursor, or Antigravity conformance. Offline parsers and refused spawn are not live receipts.
- We do not claim that the archived 2026-08-24 live demonstration certifies the current tree or sits anywhere on the evidence ladder.
- We do not claim exactly-once physical side effects. Ambiguous remote writes stay UNKNOWN until identity-exact adoption.
- We do not claim that the two TLC models prove deadlock freedom, at-most-once effects beyond their guards, or any property of the product.
- We do not claim that evolutionary search, PostgreSQL, or remote runners exist beyond design documents and refused scaffolds.
- We do not claim Jankurai, SBOM, or supply-chain release quality.
- We do not claim that a green `just fast` or a green crate test is product readiness.
- We do not claim that finishing this preprint, or writing the product-gap register, closed G1–G15.

If a sentence in marketing copy contradicts this list, the list wins.

XVII. RELATED WORK

Transactions and effects. Transaction processing is the ancestor [53], [95]: identity, isolation, and recovery under crash. Sagas name why a long-running remote effect cannot be a local commit [67]. Fencing tokens—a monotonically increasing number that the storage service itself must check—are Kleppmann’s answer to a lock holder that pauses and resumes after expiry [63], [96]; Bullet Farm’s never-reused fence, checked inside the ledger transaction and modelled in LeaseFence, is that idea applied to repository writes and remote effects. Idempotency keys make a retried request return the original result instead of a second effect [64], [65]; the #5511 class is exactly a missing idempotency key on graph materialization. Wound-wait is the classic deadlock-avoidance rule the historical adjudication adopted for multi-resource scope expansion and the models do not yet encode [94]. Confused-deputy and capability discipline are why a portal, MCP server, or vendor sandbox must not inherit forge credentials [68].

Tokens and canonical form. PASETO v4.public is the signed-token format the wire uses, chosen over JOSE for its versioned, algorithm-fixed design [60]; RFC 8785 gives the canonical JSON that makes signed payloads byte-stable across Rust and TypeScript consumers [59].

Sandboxing. Landlock restricts ambient filesystem and network rights for unprivileged processes [74]; seccomp filters system calls [75]; user, mount, PID, and network namespaces isolate process views [76]; bubblewrap composes those into an unprivileged sandbox and is what Omnigent’s native Linux wrappers use [15], [97]. Bullet Farm’s committed policy names `linux-landlock-seccomp-cgroup-v1` as its production containment reference [71]; only the network-namespace egress component exists today.

Merge queues. The “not rocket science” rule—test the exact merge result before it becomes the branch—is the origin of Bors and its descendants [44], [98], [99]; Gas Town’s Refinery is a bisecting queue of that lineage. Bullet Farm keeps the rule and moves the authority: the forge’s required checks decide, an attestor distinct from the delivery broker publishes them, and Evidence is bound to the exact Candidate rather than to a branch. Jujutsu shows why a stable Change identity matters and why a Git trailer is not enough [55]. in-toto and SLSA are the supply-chain vocabulary for the release artifacts Gas City already ships and Bullet Farm does not [100], [101].

Formal methods. The two models are TLA+ specifications checked with TLC [79], [80]; Section VII states their bounds.

Search and selection. The evolutionary design draws on quality-diversity archives and MAP-Elites [86], [87], on successive halving, Hyperband, and ASHA for staged budgets and early pruning [88]–[90], on island models for sparse recipe migration [91], and on contextual bandits with unbiased offline evaluation for the post-V1 routing learner [92], [93]. The family’s addition is not an algorithm; it is the rule that hard constraints filter before any objective and that nothing in the genome can touch authority.

Coding agents. SWE-bench is the dominant issue-resolution benchmark [33]; SWE-agent, Agentless, Open-

Hands, and AutoCodeRover optimize an agent–computer interface or a no-agent scaffold for it [35]–[38]. Multi-agent role graphs (ChatDev, MetaGPT, AutoGen) assign personas and conversation patterns, not incarnation-fenced write authority [102]–[104]. Indirect prompt injection through retrieved or repository-resident instructions is a documented confused-deputy class [66]. ACP and MCP are transport and tool seams [26], [32]. Gas Town, DeepSeek Harness, and Omnigent are the nearest *product* neighbors and are treated as such in Sections III and XII, not as straw men. None of those systems binds Evidence to an exact Candidate OID.

XVIII. LIMITATIONS

Bullet Farm is pre-release [23], [82]. There is no connected TRANSACTION_PROOF at HEAD, so every composition claim in this paper is design. Live provider, Jeryu, and GitHub gates are operator-blocked. Hub-only installation is refused because the checked-in lock is schema 2. Security quality is below the V1 bar (hub Jankurai 58/raw58 with 10 caps and 44 findings: 29 high/hard and 15 medium/soft, at the 2026-08-25 audit [23]; the figure moves with every audit). Eleven of fifteen Portal surfaces stay explicit UNKNOWN. Evolutionary multi-agent search is a normative document, not a running optimizer [56]. C2, C3, C4, C5, C6, C6b, C9–C12 are designed or out of V1; only C7 is a complete formal-methods receipt, and Section VII lists the weaknesses of that receipt. The kernel test suite on the working tree reports three skipped tests; the committed receipt counts exclude them, and the family’s no-skip-green rule applies to product profiles, not to those unit lanes [25]. PostgreSQL, remote runners, and cross-repository sagas are scaffolds or absences. This preprint’s authors are placeholders. External pins were re-verified by one reviewer on one date; Gas City’s v1.4.1 release-note artifact list could not be re-fetched and is taken from the dated snapshot. Comparison cells for DeepSeek and Omnigent Git internals that are not stated in their public architecture are marked unknown rather than inferred from plugin names. The inventory HEADs are a dated snapshot; the working tree that compiled this PDF may be newer.

XIX. CONCLUSION

The coding-agent industry is composing loops. Bullet Farm is a transaction processor for the thing those loops actually mutate: repositories and forges, under crash, timeout, and adversarial concurrency. Gas Town showed that roles, convoys, and a merge queue are valuable and that tmux, worktrees, and inferred completion are a ceiling. DeepSeek Harness showed that a session log and a plugin kernel make a single agent inspectable and recomposable. Omnigent showed that OS isolation, credential proxying, and cross-harness policy can be layered on top of the agents people already run. Bullet Farm keeps those product lessons and changes the substrate: private clones, incarnation fences, exact Candidates, independent Evidence, brokered effects, and a portal that would rather show UNKNOWN than a false empty success.

The components that exist today already enforce pieces of that contract and refuse the rest. The connected product is not done until the receipt table says so.

APPENDIX A OBSERVATION PINS

All moving-repository observations in this paper are dated 2026-08-25 UTC. Issue titles, states, and dates were re-fetched from GitHub on that date; release commits were re-fetched where the page rendered.

- Gas Town: release v1.2.1 (2026-06-06), peeled commit 319d33a91b2deca59bba6dd26be6b9daf8eaacf6. README facts used: tmux 3.0+; bd 0.57.0+; Dolt; ICU4C; “Zombie: Dead tmux session”.
- Gas Town issues: #4094 (open, 2026-05-21, v1.1.0); #3584 (closed, opened 2026-04-10); #4737 (open, 2026-08-19, v1.1.0); #4091 (open, 2026-05-21).
- Gas City: release v1.4.1, peeled commit 58ef17e3bd685fd5cf7f21286277b208d3324590 (commit verified; release notes not re-fetched); development main 2cd07e018bf3680d24b037b509e6a4bad5e623ba as pinned by the snapshot. Issues: #1181 (open, 2026-04-24, priority/p1); #5511 (open, 2026-08-22).
- DeepSeek Harness: tag dsh-v0.1.1-rc.2 (2026-08-21), commit b150a551b8d465e31e418e1b2eaf5e79bbb7d28e; docs/architecture.md at that commit.
- Omnigent: release v0.10.0, commit 40755dd8dddb07e1eb6e4055d1d9936e184ceb9b; development main 00bbbed5d92d628f975053e82dbfd1ecd542d9b20 as pinned by the snapshot; Databricks announcement 2026-06-13.
- Provider binaries (local, environment-cleared -version/-help, SHA-256 recorded in ADR 0001): Claude Code 2.1.243; Codex CLI 0.149.1; Cursor Agent 2026.08.11-e8db854; agy 1.1.19.
- Bullet Farm family subjects used for the inventory: hub 0cc7eecffd4a6fe51c0288ea4d80e02729f79848 (code/report subject) under implementation head 6eb002b8244f1251dec1cb3ce1875bb6cfcef3a4 (documentation only); kernel 0346fd531e231a88e5b1eb3a8257be04f235068b; Bullet-Git 236f4efe2b8243b3c4dd5da13269e45119b061df; portal 8272844e44d89ea3b26a5b3c4f25909a069e101 [24].

A later comparison must record a new tag or commit and date. Silently tracking main is forbidden [19].

APPENDIX B REPRODUCIBILITY

Every number in this paper comes from one of the commands below, run read-only from the canonical ordinary-clone family on 2026-08-25 at the subjects in Appendix A (hub at 6eb002b). Observed outputs are given so a reader can tell drift from error. cargo is the pinned 1.97.1 toolchain

unless stated; bullet-family is cargo run -locked -quiet -bin bullet-family - from the hub.

- cargo test -locked -p bullet-wire (hub): 83 passed, 0 failed, 0 ignored across 13 test binaries (authority_contract 9, authority_reservation 7, canonical_hostile 5, formal_traces 2, golden 5, ipc 7, ipc_client 4, launch_grant 15, policy_registry 10, policy_vlalpha2 11, validation 6, library 2).
- cargo nextest run -workspace -locked (kernel): 411 passed, 3 skipped on a working tree that also held one uncommitted lane; the committed receipt for kernel d388733/0346fd5 is 410/410 [24].
- cargo nextest run -workspace -locked (BulletGit 236f4ef): 123 passed, 0 skipped.
- npm test - --run (portal 8272844): 67 passed in 12 files.
- bullet-family check release -json (hub 6eb002b): exit 3; 26 gates, all BLOCKED: release.backup-restore, checksums, fault-suite, forge.github-app, forge.jeryu, installable-lock, installer-twice, jankurai-90, manifest-non-circular, package-matrix, platform-containment, provenance, provider.antigravity, provider.claude, provider.codex, provider.cursor, receipt-contracts, rust-msrv-1-95, rust-pinned-1-97-1, sbom, scan.dependency, scan.license, scan.secret, scan.workflow, signatures, transaction-demo.
- bullet-family doctor -json: BLOCKED; the checked-in family.lock is schema 2 and its member OIDs (kernel d174334, BulletGit ac44d74, portal 043bc72) differ from the inventory HEADs by design.
- bash formal/model-check.sh (hub): downloads and pins TLC 1.7.4 (SHA-256 936a2620...); Java 21; seed 20260824; one worker; LeaseFence 529,851 generated / 128,913 distinct / depth 24; EffectCheck 1,585 / 378 / depth 18; “2/2 models match pinned tool, source, config, and state counts.”
- policy/vlalpha1/policy.json: schema_version=vlalpha1, policy_generation=1, live_admission_enabled=false, evolutionary_authority=false, maximum_lease_ttl_seconds=15, network_default=deny, one release-signing issuer key with no audiences.
- Archived receipt .17-bundle/demo-live-receipt-live-L7b-140403.json: the fields quoted in Section X.
- Gas Town, Gas City, DeepSeek Harness, and Omnigent facts: the URLs in Appendix A and in the bibliography, fetched 2026-08-25.

The PDF is compiled with pdflatex, bibtex, and two further pdflatex passes from docs/paper/. A clean compile is not a family release subject.

APPENDIX C RED-TEAM SELF-REVIEW

A hostile reviewer will run the following attacks. Each is answered or conceded; the residual is stated.

- 1) “*Your Gas Town autopsy cites issues that do not say what you say.*” Conceded for one citation in earlier drafts, which attributed a respawn storm and a “687 respawns in 24 h” figure to Gas Town issue 742; that issue is about a settings file written to the wrong directory, and the figure has no primary source. This revision removed both and replaced them with issues 3584, 4737, and 4091, whose titles, states, and dates were re-fetched (Appendix A). Residual: issue trackers move; the class survives even if any single report is fixed.
- 2) “*Your implemented row is empty, so the architecture is vapor.*” Conceded for the connected product: nothing is above COMPONENT_PROOF. Rejected for named component receipts: Table IV and Appendix B are the contract with the reader, replayable at named commits.
- 3) “*Two TLA+ models prove the system.*” Conceded. They prove two protocols under small bounds, safety-only, with deadlock checking off, one weak invariant, one unchecked invariant, and no wound-wait (Section VII). They do not prove the product.
- 4) “*You had a live end-to-end run and deleted it; either it worked or it did not.*” Both. The archived receipt shows the seven-clause transaction running once on real processes; the admission mechanism it ran under was an environment token that the family later rejected as not being admission, so the surface was removed and its replacement (signed grants, egress receipt, ratified policy) is not yet complete. Residual: the paper reports the receipt as [ARCHIVED], outside the ladder.
- 5) “*Providers propose is unenforceable: the provider’s own sandbox can write files.*” Partly conceded. The provider workspace generation is private and its bytes are never applied; only a validated PatchProposal reaches bullet-gitd. But S1/S2 containment is designed, only the network dimension is a component, and no transaction has yet composed admission, proposal, and apply. Residual: stated in Section VI.
- 6) “*A single-user loopback threat model is trivial.*” Rejected as trivial, conceded as narrow. The adversaries are the model, a concurrent incarnation, a retrying runner, hostile repository content, and a lying projection—the ones the issue corpus shows cause loss. Multi-tenant and remote-runner threats are named out of scope, not implied.
- 7) “*You ship a security paper with 44 findings: 29 high/hard and 15 medium/soft.*” Conceded. The figure is reported, dated, and blocks release by the family’s own rule; the paper claims no security quality.
- 8) “*Omnigent’s sandbox plus credential proxy is stricter than your V1.*” Conceded for OS-isolation maturity.

Rejected as write-authority: the wrapped harness still runs Git and a policy cannot invalidate stale evidence.

- 9) “*The bindings are just fencing tokens, idempotency keys, and merge queues with new names.*” Largely conceded, and cited as such in Section XVII. The claimed contribution is the composition over repository subjects—fence at the effects gateway, Candidate identity from base/head/tree/patch, Evidence invalidated by subject change, preservation before destruction—and the refusal to let any model output cross those lines. Whether that composition is worth its machinery is a benchmark question this paper does not answer.
- 10) “*An internal score or a green crate test is being sold as a result.*” Rejected. No internal weighted score appears; the only scores quoted are the historical adjudication’s own (96.4/100 for the paper architecture, about 0.5/10 for implementation), reported as provenance. Every test count carries its subject and class.

Further residuals accepted from earlier reviews: the eleven UNKNOWN portal surfaces are the correct projection of missing farmd routes, not unfinished UI; v1alpha2 on HEAD does not mean live is on, because the Kernel loader accepts v1alpha1 only and the committed policy is generation 1; the inventory HEADs are yesterday’s tips for any later commit; the same-UID swap after verification is an open install hole (G1); #5511 may be fixed upstream while the class remains; closing this preprint closed nothing in G1–G15; and a worktree with a unique branch is not isolation because .git, hooks, and the index are shared.

APPENDIX D REMAINING PRODUCT GAPS

Table VIII is the operator-facing remainder. The executable form is bullet-family check release -json (26 static negative gates). The prose form is the hub product-gap register, which maps every release.*id onto G1–G15 [82]. This appendix exists so a red team cannot say the paper hid an open gate behind architecture.

Operator-only closers, which no agent may flip: ratify ADR 0012 with a real runner key; restore scoped Jeru authentication on an unmodified forge; publish schema-3 lock inputs; configure a GitHub App test repository. Documentation that waits on typed commands—upgrade/rollback, signer rotation, freeze-enforced cards—is deferred on purpose. Inventing those runbooks now would invent a CLI.

V1 is done when every closure-plan slice V1-S0..V1-S8 has a current independently verifiable receipt from the same signed subjects [24]. “Agents ran” is not that receipt.

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TABLE VIII
REMAINING V1 PRODUCT GAPS (2026-08-25). NONE OF THESE IS CLOSED BY THIS COMPILE.

ID	Gap	Why it is still open	What already exists
G1	Hub-only signed install	Schema-2 lock refused on purpose	Descriptor-relative setup fixture
G2	Five-plane transaction	No signed TRANSACTION_PROOF	Atomic lease/command; fixture E2
G3	Production Kernel writes	No signed lease transport / JSON-RPC dispatch	DB-clock leases; launch grant
G4	Production BulletGit writes	Public clone fail-closed	Dissociate, hostile-git, preservation
G5	Live provider conformance	Committed policy live-false	Four offline parsers; egress probes
G6	Jeryu live effect	No authenticated read-back	Local bare-forge component
G7	GitHub live effect	No App-test-repo receipt	Specified adapter
G8	Security release floor	Jankurai 58/raw58; 10 caps; 44 findings: 29 high/hard + 15 medium/soft	Fail-closed local scan
G9	Signed five-target release	No builder / SBOM / protected signer	Verify+extract of a signed archive
G10	Non-Linux containment	Mutation refused off Linux	Linux isolation reference
G11	Evolutionary runtime	Recipes are design-only	Normative evolutionary document
G12	Family check release	26/26 gates BLOCKED	Negative inventory
G13	Portal product surfaces	11 of 15 surfaces explicit UNKNOWN	Four projected surfaces; CSRF/202
G14	farmd production API	Snapshot/SSE subset, not the designed route catalog	Authenticated command ingress
G15	Cognitive persistence	Tasks/roles are IDs or design	Wire shapes; offline parsers

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